DAURES GREEN HYDROGEN CONSORTIUM
DAURES GREEN HYDROGEN VILLAGE
University of Stuttgart
IER Institute of Energy Economics and Rational Energy Use
With funding from the:
Federal Ministry of Research, Technology and Space

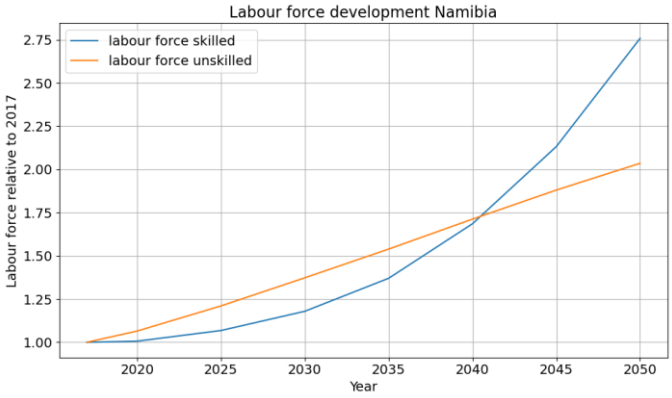
Macroeconomic scenario framework

General assumptions

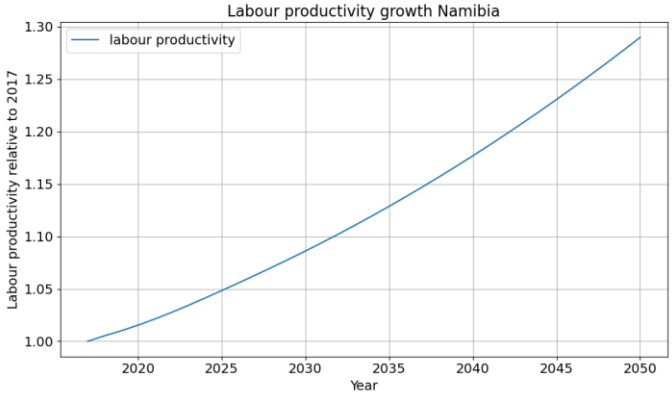
To conduct a scenario analysis a scenario framework needs to be defined. Results analysis is always relative to a reference scenario. The expected effects of policy measures can then be assessed through counterfactual scenarios.

For the definition of a reference scenario, some assumptions need to be made:

Labour force development:



Labour productivity growth:



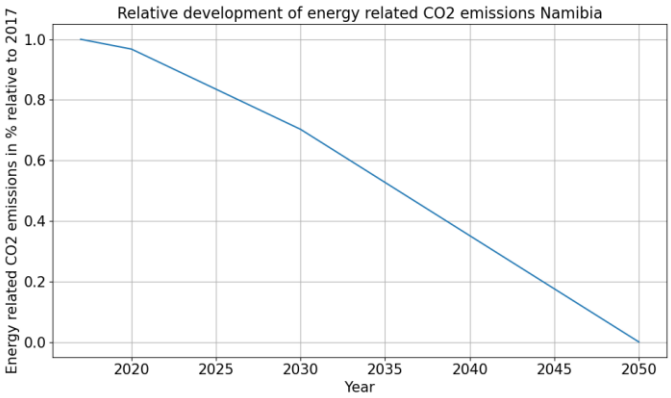
In addition, assumptions are made with respect to **elasticities of substitution**, **technology costs** and costs associated with **technology expansion**.

E-fuels and **synthetic gas** can only be made from domestically produced green hydrogen and clean carbon sourced from the agriculture sector. **Electricity** from the grid at Namibian power prices is used for electrolysis. **Electricity generation technologies** that are inactive in the model base year 2017 are also unavailable further on. The only exception is hydrogen gas turbine.

Climate policies

The reference scenario is an emission reduction scenario. Only energy related CO2 emissions are considered. For all regions except Namibia the emission reductions are in accordance to the "Announced Pledges" of the IEA WEO 2024 report. For Namibia, the NDC 2023 "mitigation" scenario is assumed until 2030. After 2030 an emission reduction pathway in line with Net-Zero ambitions for 2050 is assumed. No further policies are considered.

CO2 emissions pathway:



Counterfactual scenarios

For the comparative scenario analysis, the reference scenario "Net Zero" is compared to two counterfactual scenarios. Climate policy assumptions (the emissions reduction pathway) are equal in all three scenarios. However, in the two counterfactual scenarios hydrogen production is fostered through a subsidy on green hydrogen production. In the first counterfactual scenario "External Subsidy" this subsidy is financed through foreign governments, whereas in the "Domestic Subsidy" scenario, the Namibian government has to pay the subsidy to the hydrogen producers.

Within the CGE modeling framework, subsidizing a specific technology by 1.0 means that 100% of the current market price of the energy carrier is paid as a subsidy from the government (domestic or foreign) to the producers on top of the payment received from the sale of the energy carrier.

Scenario	Net Zero	External Subsidy	Domestic Subsidy
Climate policy	World: IEA WEO 2024 „Announced pledges"; Namibia: NDC scenario		
Hydrogen subsidy	-	3.0	3.0
Subsidy financing	-	Foreign Government	Namibian Government

Contact

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https://github.com/IER-Hy4Daures/Fact-Sheets